Learning to Adapt Using Test-Time Images for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Current methods for salient object detection in optical remote sensing images (RSI-SOD) adhere strictly to the conventional supervised train–test paradigm, where models remain fixed after training and are directly applied to test samples. However, this paradigm faces significant challenges in adapting to test-time images due to the inherent variability in remote sensing scenes. Salient objects exhibit considerable differences in size, type, and topology across RSIs, complicating accurate localization in unseen test images. Moreover, the acquisition of RSIs is highly susceptible to atmospheric conditions, often leading to degraded image quality and a notable domain shift between training and testing phases. In this work, we explore test-time model adaptation for RSI-SOD and introduce a novel multitask collaboration approach to tackle these challenges. Our approach integrates a self-supervised auxiliary task, specifically image reconstruction, with the primary supervised task of saliency prediction to achieve collaborative learning. This is accomplished through an architecture that comprises a shared feature encoder and two distinct task-specific decoders. Most importantly, the self-supervised image reconstruction task optimizes model parameters using unlabeled test-time images, allowing adaptation to test distributions and enabling flexibly scene-dependent representation learning. In addition, we design a cross-task modulation module (CMM) positioned between the task-specific decoders, which fully exploits intertask correlations to enhance the adjustment of saliency representations. Extensive experimental evaluations confirm the superiority of our method across three widely used RSI-SOD benchmarks and validate the robustness of our proposed test-time adaptation strategy against diverse types of RSI corruptions.**

***Index Terms*— Optical remote sensing images, salient object detection (SOD), self-supervised learning (SSL), test-time adaptation.**

I. Introduction

SALIENT object detection (SOD) is a fundamental task in computer vision, which targets at detecting the most conspicuous objects in images. It promotes various vision applications, including image retrieval [1], photo editing [2], image compression [3], and object recognition [4]. Compared with SOD in natural scene image (NSI-SOD), SOD in optical remote sensing image (RSI-SOD) has emerged as a new and more challenging research topic. RSIs are acquired by sensors mounted on airplanes or satellites, covering a wide range of view about the Earth surface. Moreover, the acquisition process is easily affected by various imaging conditions (e.g., shooting altitude, atmospheric interference) and with minimal human intervention. The distinctive characteristics of RSIs pose specific challenges for RSI-SOD, e.g., varying object scales and orientations, diverse texture, and background patterns.

Deep neural-network-based methods [5], [6], [7], [8] can learn discriminative representation from labeled RSIs and generalize to challenging cases. The majority of works focus on designing multiscale feature integration strategies to address the issue of scale variation [5], [6], [9]. The pioneer method, LVNet [5], extracts multiscale representations from multiresolution inputs and uses a nested structure to gradually fuse scale-aware representations from deep to shallow layers. Besides multiscale feature integration, long-range context modeling establishes the global relationship dependencies and can eliminate the interference of cluttered background [7], [8], [10]. For example, DAFNet [7] incorporates a global context-aware attention mechanism and a dense fluid structure to propagate relationships from deep to shallow layers. MCCNet [10] integrates global image-level features and several type of attention modules, e.g., foreground and background attention, to derive discriminative object representations. To obtain saliency objects with well-defined structures and boundaries, some works use edge and skeleton information to further refine salient object regions [11], [12], [13]. EMFINet [11] learns edge feature representations under the explicit supervision of salient object edges and integrates them to refine multiscale network feature representations. ERPNet [12] uses edge information to sharpen the position cues and recurrently shift the focus of decoding stages to salient regions.

Despite considerable advancements in RSI-SOD, current methods still cannot adequately adapt to the variability and diversity of RSIs in different testing scenarios. The conventional train–test paradigm, in which models remain fixed after training and are directly applied to test images, may struggle with test-time adaptability. This is attributed to the assumption of independent and identically distribution between the train

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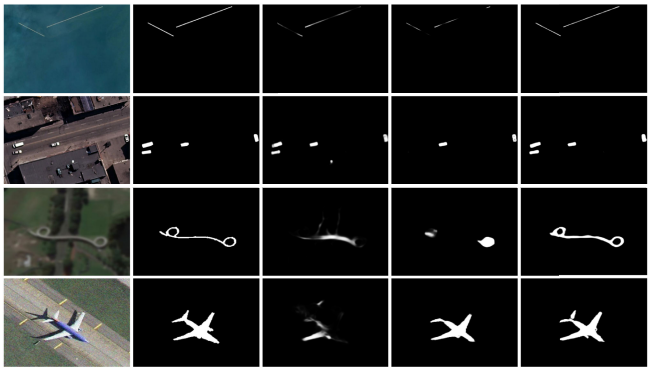


Fig. 1. Visual comparison of RSI-SOD methods across diverse test scenarios. (a) Optical RSIs. (b) GT. (c) DAFNet [7]. (d) ACCoNet [21]. (e) Our proposed method. The top two rows depict scenarios featuring irregular structures (the first row) and multiple small objects (the second row). The bottom two rows present corrupted RSIs affected by blur (the third row) and impulse noise (the fourth row).

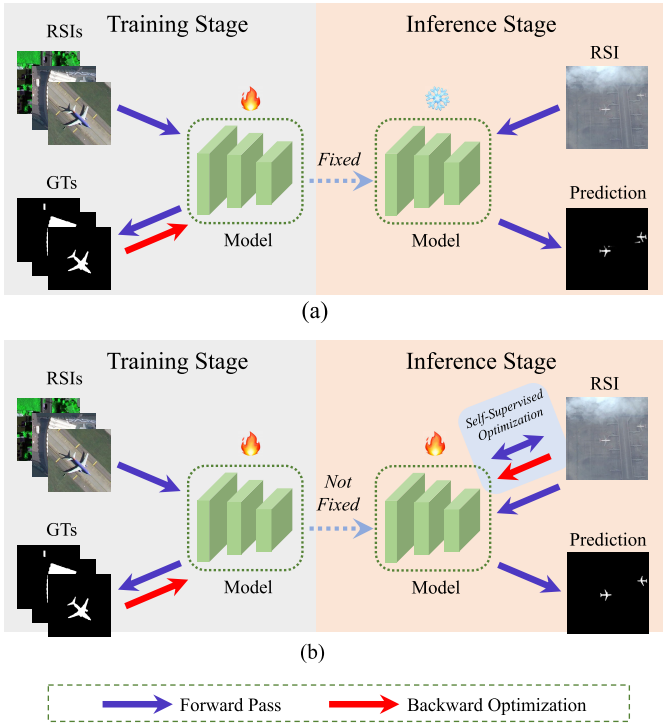


Fig. 2. Illustrative comparison of our approach with the existing RSI-SOD methods. (a) Existing methods. (b) Our method.

and test samples. The lack of test-time adaptability may hinder the application of these models in real-world circumstances. The characteristics of salient objects in RSIs are marked by large variability in several aspects such as size, type, topology, and orientation, making it challenging to precisely localize them in various test scenarios (as shown in the first two rows of Fig. 1). In addition, the acquisition of remote sensing images is highly susceptible to atmospheric conditions, leading to degraded image quality and a notable domain shift between training and testing phases (as shown in the bottom two rows of Fig. 1). As a result, models that are tuned exclusively on a static training dataset struggle to adapt to various testing scenarios.

To address these challenges, we draw inspiration from the idea that the unlabeled test RSIs provide a clue about the distribution from which they originate. Our approach uses a test-time adaptation strategy that fine-tunes network parameters during test time and improves final inference accuracy. Fig. 2 provides a schematic comparison between the existing RSI-SOD methods and our proposed method. The unlabeled test RSIs cannot directly contribute to model optimization in a supervised framework. However, they can facilitate model adjustment through self-supervised learning (SSL) techniques [14], [15], [16], [17]. Recent progresses in SSL use various signals for supervision, such as rotation angles [14], patch indexes [18], clustered pseudolabels [19], and instance-level labels [16]. These methods typically rely on image-level or region-level signals, which are suboptimal for the RSI-SOD task that requires discriminant pixel-level representations. Masked image modeling (MIM) [17], [20] obscures portions of image patches and leverages the visible segments to predict the obscured pixel contents. Yet, the effectiveness of MIM in learning reliable pixel-level representations is limited, particularly for RSIs that encompass a broad view and may contain highly varied possibilities within certain masked regions. In this work, we turn to a straightforward and intuitive self-supervised task, i.e., image reconstruction. This task can learn informative location-aware representations from the data and further collaborate with saliency prediction task. By integrating image reconstruction as an auxiliary task, we effectively tailor the model's responses to the specific nuances of RSIs during test time, improving the robustness to diverse test scenarios.

In this article, we propose a novel multitask collaboration approach to address the complex challenges encountered in diverse test-time scenarios. A network architecture is proposed, featuring a shared feature extraction encoder and two distinct substream decoders dedicated to saliency prediction and image reconstruction, respectively. Moreover, we design a cross-task modulation module (CMM) that uses intertask correlations to improve the adjustment of saliency representations. During training, the entire architecture, including the shared encoder, both substream decoders, and CMM, is optimized using both saliency and reconstruction losses in a standard supervised fashion. Prior to inference, the model undergoes a test-time training (TTT) phase where the shared encoder and reconstruction decoder are fine-tuned by reconstruction loss. Finally, at the inference stage, the updated model generates the final saliency predictions by further using intertask correlations through CMM. As depicted in Fig. 1, our method demonstrates robust performance across a variety of challenging test scenarios, such as irregular structures, multiple small objects, blur corruption, and impulse noise corruption. Through extensive experiments on three public RSI-SOD benchmarks, the efficacy of our proposed method is well-examined and justified. Furthermore, our method has been validated on several types of image corruptions, affirming the robustness of the proposed test-time adaptation strategy. The contributions of this work are summarized as follows.

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1) We propose a test-time adaptation method for RSI-SOD to address the variability of RSIs across unseen test-time scenarios.
2) We design a multitask collaboration architecture that integrates a self-supervised auxiliary task, specifically image reconstruction, to adapt to unlabeled test images and support saliency detection.
3) We develop a CMM that exploits intertask correlations to improve the adjustment of saliency representations.

The rest of this article is organized as follows: Section II provides a review of related works. Section III elaborates on our proposed network architecture. In Section IV, we describe the optimization process during both training and TTT stages. Extensive experiments and ablation studies are presented in Section V. Finally, conclusions are drawn in Section VI.

II. Related Works

A. SOD in Natural Scene Images

NSI-SOD has been extensively researched and witnessed remarkable progress in recent decades. Early methods focused on exploiting saliency priors by designing hand-crafted descriptors, such as color contrast [22], [23], frequency residual [24], [25], statistic priors [26], and background priors [27]. In addition, some methods proposed to learn the mapping from hand-crafted image descriptors to saliency maps in a supervised manner [28], [29].

Empowered by the representation capability of deep neural networks, researchers have constructed end-to-end network architectures to learn salient object representations. Neural-network-based methods investigated several aspects to promote SOD, including multiscale feature integration [30], [31], global dependency modeling [32], [33], edge guidance [34], [35], and high-level feature guidance [32], [36]. Multiscale feature integration explores the complementarities between multiple network levels to enhance detection robustness for varying scale objects. Global dependency modeling aims to pop out salient regions using the global context-aware and comprehensive relationships. Object edge serves as a valuable clue for enhancing salient object regions and have been used to refine the salient regions more uniformly and completely. High-level feature learning provides a comprehensive understanding of the whole image, which can guide salient object extraction more effectively.

Due to the distinct characteristics between natural scene images and optical remote sensing images, RSI-SOD poses specific challenges, including highly intricate topologies and varying orientations. Previous studies [10], [21] suggest that directly applying existing NSI-SOD methods to RSI-SOD is ineffective due to the intrinsic properties of RSIs. Consequently, researchers have recently begun developing specialized methods to promote RSI-SOD and its applications.

B. SOD in Remote Sensing Images

RSI-SOD has received a growing interest from the research community in recent years. Traditional methods rely on exploiting predefined priors to obtain discriminative RSI descriptors, such as texture and color contrast [37], spectral energy [38], and background priors [39]. For instance, Ma et al. [37] integrated the representation of texture saliency and color saliency into a pixel-level saliency map. Zhao et al. [39] used background cues and designed a sparsity-guided method for saliency detection.

Inspired by the success of CNNs in NSI-SOD and various vision tasks, researchers have developed a number of CNN-based methods tailored for RSI-SOD. Early efforts have been devoted to extracting comprehensive feature representations from multiresolution inputs [5], [11], explicitly tackling the scale variability issue in RSI-SOD. Subsequently, interactions between multiple network levels of a CNN architecture have also been explored to improve the robustness of detecting salient object of varying sizes [6], [21], [40]. A line of methods proposed to use attention mechanism to model the global contextual dependencies, eliminating the interference of background clutter in RSIs [7], [10], [41]. For instance, Zhang et al. [7] used self-attention mechanism to capture long-range context dependencies and propagated them from shallow to deep layers in a cascaded manner. Moreover, exploring the interaction between global and local contextual information is crucial for enhancing salient object details [42], [43], [44]. For instance, Li et al. [42] adopt a global-to-local paradigm, using transformer to extract embeddings with long-range dependencies, while incorporating direction-aware attention to refine local details. Coarse object boundary and incomplete salient region severely affect the quality and application of salient object maps, hindering their usage in real-world applications. To mitigate this issue, several methods have incorporated object edge cues to guide the refinement of salient object regions [12], [45].

Despite remarkable progress that has been achieved, the diversity and variability still pose serious challenges to RSI-SOD. The existing methods adopt the conventional train–test framework, in which models are fixed after training and directly applied for inference. In contrast, we propose to leverage TTT to fully use the clues provided by test samples. The model adjustment that is subsequent to training and prior to inference can effectively learn test-sample-related distribution and cope with domain shift.

C. Test-Time Adaptation

Test-time adaptation aims to adapt models to generalize to unseen test data during inference. Sun et al. [46] proposed a rotation-based self-supervised method for improving model's prediction performance when training and testing data come from different distributions. Wang et al. [47] proposed to adapt models by test entropy minimization and optimize channelwise affine transformations. Liang et al. [48] proposed to learn target-specific feature representations by aligning the target domains to the source hypothesis using information maximization and self-supervised pseudolabeling. Gandelsman et al. [49] used the masked patch prediction as the pretext tasks to adapt the network parameters. Chen et al. [50] leveraged self-supervised contrastive learning to facilitate test-time feature learning and further used an online pseudolabeling

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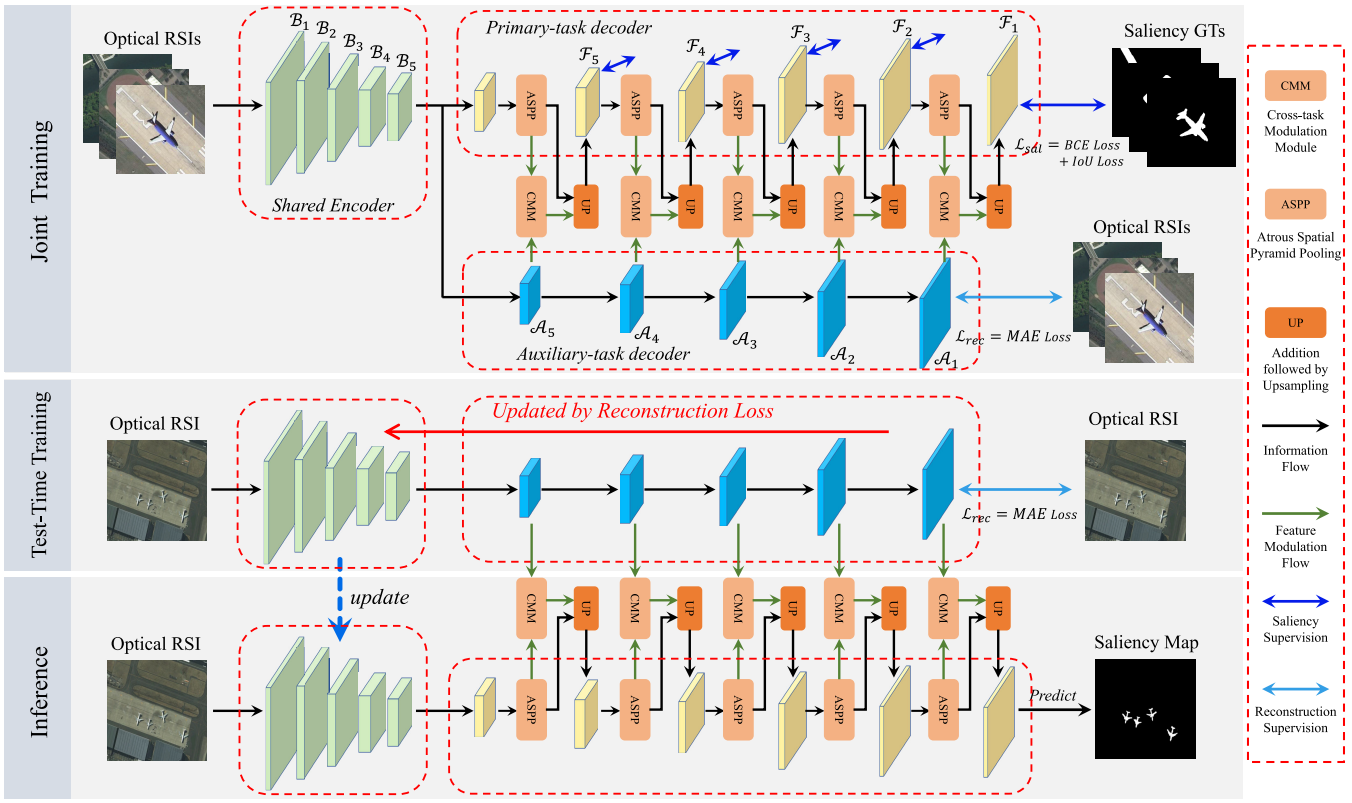


Fig. 3. Overall pipeline of our proposed TTT methodology.

scheme to refine the learning process. Besides image recognition, the test-time adaptation is extended to more vision tasks, such as video frame interpolation [51], video action recognition [52], 3-D point cloud recognition [53], and 3-D human mesh reconstruction [54]. The above test-time adaption methods achieve impressive performance across diverse testing scenarios. Therefore, we introduce the test-time adaptation into RSI-SOD task to address the variability across test RSI scenarios. To tackle the specificity of RSI-SOD, we propose to use a self-supervised pixel-level learning task, i.e., image reconstruction, as the pretext task. And we further entangle the pretext task with the primary task via a CMM to enable better task collaboration.

III. Methodology

The proposed network architecture and the corresponding workflow are schematically illustrated in Fig. 3. The workflow delineates three distinct phases: training, TTT, and inference. The training phase uses a standard supervised learning framework, tailored to achieve a multitask learning objective. Subsequent to training and prior to inference, TTT is implemented, wherein the model's parameters, including those of the shared encoder and auxiliary-task decoder, are fine-tuned in accordance with the test optical RSI inputs. The final inference stage makes the saliency predictions using the adjusted model parameters.

The network's architecture is composed of a shared feature extraction encoder, a primary-task decoder, an auxiliary-task decoder, and a CMM. The shared encoder is dedicated to extracting base feature representations from the input RSIs, which are then fed into the dual decoders. The dual decoders concurrently address multiple objectives: SOD and image reconstruction. Notably, the network incorporates a CMM, designed to foster interaction between the primary- and auxiliary-task decoders.

We adopt VGG-16 [55] as the shared encoder to extract multiscale base feature representations. Suppose we have an input RSI $\mathcal{I} \in \mathbb{R}^{H\times W\times C}$, and we can obtain multilevel representations $\{\mathcal{B}_i\}_{i=1}^{5}$ from the encoder, where $\mathcal{B}_i$ has the spatial resolution $(H/2^{i-1})\times(W/2^{i-1})$, and i denotes the layer index. The encoder representations are fed into the dual decoders, i.e., the primary- and auxiliary-task decoders, to accomplish SOD and image reconstruction, respectively. The joint multitask learning facilitates comprehensive feature extraction during the training stage. Most importantly, during the test-time, the auxiliary-task decoder along with the shared encoder can be optimized via the self-supervised image reconstruction objective. The test-time update adjusts the model parameters according to the new test RSI, improving its adaptability to diverse challenging RSI scenarios.

A. Primary-Task Decoder for Saliency Prediction

The primary-task decoder aims to accomplish SOD. Following U-Net design [56], we construct a decoder architecture that is symmetric to the encoder structure. Skip connections are incorporated between the corresponding encoder and decoder layers for feature enhancement. It is noted that the skip connections are omitted in Fig. 3 for simplicity.

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Suppose the feature representations of the decoder network are denoted as $\{\mathcal{F}_i\}_{i=1}^{5}$, where i denotes the layer index. The representation from the upper level $(i+1)$ of the decoder is $2\times$ upsampled, and then concatenated with the current-level (i) encoder representation. A convolutional layer with batch normalization and ReLU is applied to the concatenated representation for feature transformation. The process can be expressed as follows:

$$\mathcal{F}_i^t = \text{Conv}(\text{Up}(\mathcal{F}_{i+1}) \,||\, \mathcal{B}_i) \tag{1}$$

where i is the current layer index, $\mathcal{F}_{i+1}$ denotes the upper level representation from the decoder, $||$ denotes concatenation along the channel dimension, Up denotes the bilinear upsampling function, and Conv denotes a convolutional layer followed by batch normalization and ReLU activation.

Feature representations extracted by convolutional layers are from limited receptive fields, lacking the capability to address varying object scales. Therefore, inspired by semantic segmentation methods [57], [58], we use atrous spatial pyramid pooling (ASPP) [57] to enlarge the receptive fields, capturing extensive contextual information within each network layer. The process is expressed as

$$\mathcal{F}_i^a = \mathcal{ASPP}\left(\mathcal{F}_i^t\right) \tag{2}$$

where $\mathcal{ASPP}$ denotes the function of the ASPP module.

Previous TTT methods [46], [48] only optimize the shared encoder part which derives multitask decoder structures. However, the collaboration between the multitask decoders lacks exploration, which may hinder further improvement for RSI-SOD. In our method, we first modulate the auxiliary task features $\mathcal{A}_i$ using $\mathcal{F}_i^a$, and then transfer the modulated representations to the primary-task part to complement its representations. The modulation process can be expressed as

$$\mathcal{F}_i = \mathcal{F}_i^a + \mathcal{MOD}\left(\mathcal{F}_i^a, \mathcal{A}_i\right) \tag{3}$$

where $\mathcal{A}_i$ is the feature representation from the auxiliary-task decoder, and $\mathcal{MOD}$ indicates the feature modulation function, which will be elaborated in Section III-C. We adopt multilevel supervision to supervise the learning of $\mathcal{F}_i$ at each decoder level. The final saliency output is predicted from the last layer of the primary-task decoder, i.e., $\mathcal{F}_1$.

B. Auxiliary-Task Decoder for Image Reconstruction

The auxiliary-task decoder aims to accomplish the image reconstruction task, which can learn extra and complementary feature representations for SOD. Similar to the primary-task decoder, we adopt U-shaped structure with skip connections as the decoder structure. Denoting the feature representations of the auxiliary task decoder as $\{\mathcal{A}\}_{i=1}^{5}$, the decoding process can be simply expressed as

$$\mathcal{A}_i = \text{Conv}(\text{Up}(\mathcal{A}_{i+1}) \,||\, \mathcal{B}_i) \tag{4}$$

where $\mathcal{A}_i$ and $\mathcal{A}_{i+1}$ denote the current-level and upper level representations of the auxiliary-task decoder, respectively, $||$ denotes concatenation along the channel dimension, up denotes bilinear upsampling, and Conv denotes a convolutional layer followed by batch normalization and ReLU activation.

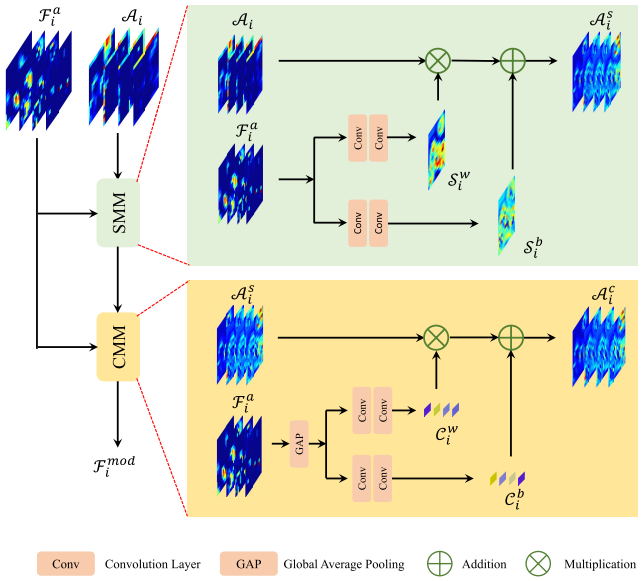


Fig. 4. Pipeline of the CMM.

A prediction layer (3×3 convolution) is appended after the last layer of the auxiliary-task decoder to produce the reconstruction output $\mathcal{R} \in \mathbb{R}^{H\times W\times C}$. In our work, we adopt the simple and standard decoding structure in the reconstruction branch to examine the effectiveness of multitask learning and test-time adaptation. More complicated structures can be incorporated for further performance improvement.

C. Cross-Task Modulation

Multitask learning with a shared encoder and dual-stream decoders (i.e., Y-structure) can facilitate more general and comprehensive feature extraction. Moreover, the auxiliary task in our method, i.e., self-supervised image reconstruction, can be used to fine-tune the shared encoder during test time. The Y-structure is widely adopted in TTT paradigm [46], [53]. However, this architecture might be suboptimal for SOD, as the separate multistream decoders do not collaborate effectively, and the fine-grained pixel-level representations in the auxiliary-task stream are not adequately exploited for the primary task. To mitigate this issue, we introduce a CMM that modulates auxiliary-task features in accordance with primary-task feature distribution and integrates them to enhance saliency prediction. Specifically, the CMM modulates auxiliary-task features along both the spatial and channel dimensions. The pipeline of the CMM is depicted in Fig. 4. The overall modulation process can be expressed as

$$\mathcal{F}_i^{\text{mod}} = \mathcal{CMM}\left(\mathcal{SMM}\left(\mathcal{A}_i, \mathcal{F}_i^a\right), \mathcal{F}_i^a\right) \tag{5}$$

where $\mathcal{SMM}(X, Y)$ and $\mathcal{CMM}(X, Y)$ denote modulating X based on Y distribution in the space and channel domains, respectively. This formula describes the function of $\mathcal{MOD}$ in 3. We explain the two types of modulations in the following.

1) Space Modulation: It targets at modulating the auxiliary-task features based on the spatial distribution of the primary-task features. The $\mathcal{F}_i^a$ is fed into two separate convolutional layers to generate a weight map $\mathcal{S}_i^w \in \mathbb{R}^{H\times W\times 1}$ and a bias

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map $\mathcal{S}_i^b \in \mathbb{R}^{H\times W\times 1}$, respectively. Next, the weight and bias maps are leveraged to modulate the auxiliary-task features, as follows:

$$\mathcal{A}_i^s = \left(\mathcal{A}_i \otimes \mathcal{S}_i^w\right) \oplus \mathcal{S}_i^b \tag{6}$$

where $\otimes$ denotes locationwise multiplication, and $\oplus$ denotes locationwise addition. The $\mathcal{A}_i^s$ is the output representation adjusted by the spatial distribution of the primary-task features.

2) Channel Modulation: It targets at modulating the auxiliary-task features based on the channel distribution of the primary-task features. Initially, an average pooling is applied to $\mathcal{F}_i^a$ to generate a global channelwise descriptor $\mathcal{G}_i \in \mathbb{R}^{1\times 1\times C}$. Next, $\mathcal{G}_i$ is fed into two separate 1×1 convolutional layers to produce a weight vector $\mathcal{C}_i^w \in \mathbb{R}^{1\times 1\times C}$ and a bias vector $\mathcal{C}_i^w \in \mathbb{R}^{1\times 1\times C}$, respectively. Finally, $\mathcal{A}_i^s$ is modulated in channel domain by the following process:

$$\mathcal{A}_i^c = \left(\mathcal{A}_i^s \otimes \mathcal{C}_i^w\right) \oplus \mathcal{C}_i^w \tag{7}$$

where $\otimes$ denotes channelwise multiplication, and $\oplus$ denotes channelwise addition. The $\mathcal{A}_i^c$ is the output representation adjusted by the channel distribution of the primary-task features.

IV. Model Optimization

A. Multitask Learning Objectives

1) Salient Object Detection: We adopt a hybrid loss function, including binary cross-entropy (BCE) and intersection over union (IoU), to optimize the SOD task. The BCE loss is calculated as

$$\mathcal{L}_{\text{bce}}(S,G) = -\sum_{i=1}^{H}\sum_{j=1}^{W}\left[G_{i,j}\log S_{i,j} + \left(1-G_{i,j}\right)\log\left(1-S_{i,j}\right)\right] \tag{8}$$

where H and W denote the height and width of the RSI, respectively. $S \in [0,1]^{H\times W}$ and $G \in \{0,1\}^{H\times W}$ are the saliency prediction and the saliency ground truth (GT), respectively. The IoU loss is calculated as

$$\mathcal{L}_{\text{iou}}(S,G) = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} S_{i,j}G_{i,j}}{\sum_{i=1}^{H}\sum_{j=1}^{W}\left(S_{i,j}+G_{i,j}-S_{i,j}G_{i,j}\right)}. \tag{9}$$

We obtain five saliency outputs $\{S^l\}_{l=1}^5$ at each network level from the primary-task decoder, which are generated by applying a prediction layer (convolution and upsampling operations) to $\mathcal{F}_i$, respectively. The overall salient object loss is calculated as

$$\mathcal{L}_{\text{sal}} = \sum_{l=1}^{5}\left[\ \mathcal{L}_{\text{bce}}\left(S^l,G\right) + \mathcal{L}_{\text{iou}}\left(S^l,G\right)\ \right] \tag{10}$$

where l denotes the layer index.

2) Image Reconstruction: We adopt mean absolute error (MAE) as the loss function to supervise the image reconstruction task. The reconstruction loss is calculated as

$$\mathcal{L}_{\text{mae}} = \sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{c=1}^{C}\left|\mathcal{R}_{i,j,c} - \mathcal{I}_{i,j,c}\right| \tag{11}$$

where $\mathcal{I}$ is the input RSI, $\mathcal{R}$ is the reconstruction output of the auxiliary-task decoder, and H, W, and C denote the height, width, and channel of the RSI, respectively.

B. Optimization During Training

During training, the proposed entire architecture is optimized in an end-to-end manner. We optimize the primary and auxiliary tasks simultaneously by a multitask learning objective, which is expressed as follows:

$$\mathcal{L}_{\text{training}} = \mathcal{L}_{\text{sal}} + \gamma * \mathcal{L}_{\text{mae}} \tag{12}$$

where γ is the hyperparameter that controls the ratio of the reconstruction loss. We empirically set γ to 0.1 in our experiment.

C. Optimization During TTT

Subsequent to training and prior to inference, we conduct TTT, wherein the model's parameters are fine-tuned based on the new test RSIs. Specifically, we optimize the parameters of the shared encoder and auxiliary-task decoder via the image reconstruction loss. The learning objective can be expressed as follows:

$$\mathcal{L}_{\text{test-time}} = \mathcal{L}_{\text{mae}}. \tag{13}$$

The TTT enables the network to adjust its parameters to fit the specific test samples. Finally, at inference, the input is forwarded through the updated encoder and primary-task decoder to generate saliency maps. Most importantly, the representations from the auxiliary-task decoder are modulated and integrated into the primary-task decoder, benefiting the final saliency predictions.

V. Experimental Evaluation

A. Experimental Protocol

1) Datasets: We conduct experimental evaluation on three commonly used RSI-SOD benchmarks, including ORSSD [5], EORSSD [7], and ORSI-4199 [45]. These three datasets contain 800, 2000, and 4199 RSIs with the corresponding train–test splits and pixel-level salient objects' annotations. Furthermore, to better evaluate the robustness of the proposed method, we construct corrupted RSI datasets to mimic real-world challenges in remote sensing imagery. We use [59] to generate three typical types of corruptions, including Gaussian blur (GB), Gaussian noise (GN), and contrast reduction (CR), on copies of the test sets for EORSSD and ORSSD. Visual examples of these image corruptions are presented in Fig. 5. The corrupted datasets can be used to comprehensively benchmark the robustness of RSI-SOD methods.

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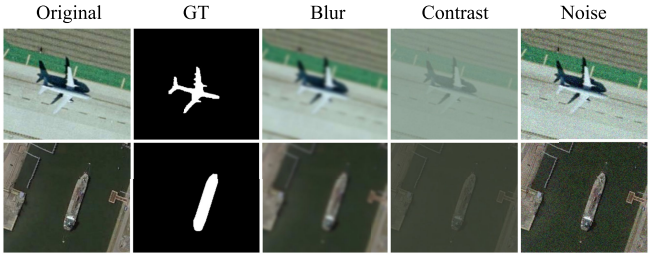


Fig. 5. Visual examples of the image corruptions, including GB, CR, and impulse GN.

2) Evaluation Metrics: We use five evaluation metrics to assess the performance of our method and compare it with the existing approaches, including F-measure [60], MAE, S-measure [61], E-measure [62], and precision–recall (PR) curve. F-measure is calculated as the weighted average of precision and recall

$$F_\beta = \frac{\left(1+\beta^2\right) \times \text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \tag{14}$$

where β^2 is set to 0.3 to assign greater weight to precision in SOD [30]. MAE measures the average absolute difference at all the pixel locations

$$\text{MAE} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \|S_{i,j} - G_{i,j}\| \tag{15}$$

where H and W denote the height and width of the input RSI, respectively. $S_{i,j}$ and $G_{i,j}$ denote the saliency prediction and the corresponding GT at pixel location (i, j). S-measure evaluates the structural similarity between the predicted salient regions and the GTs. The S-measure is computed as

$$S_m = \alpha * S_o + (1-\alpha) * S_r \tag{16}$$

where S_o is an object-aware similarity item and S_r is a region-aware similarity item, and α is set to 0.5 to assign equal contribution to both the similarities [61]. E-measure assesses the similarity between the predicted saliency maps and the GTs by simultaneously considering local and image-level saliency values. It is calculated using the alignment matrix ϕ and the enhanced alignment matrix $\theta(\phi)$ [62]

$$E_\phi = \frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} \theta(\phi). \tag{17}$$

The PR curve plots the precision rate against the recall rate by gradually adjusting the threshold of the saliency map from 0 to 255. It provides a visual representation of the tradeoff between precision and recall.

3) Implementation Details: The proposed approach is implemented using PyTorch and trained on an Nvidia RTX 3090 GPU. VGG-16 [55] is adopted as the feature extraction backbone for our proposed architecture. The network parameters are optimized using the Adam optimizer [63]. During training stage, we train our model for 40 epochs and set the initial learning rate to 1×10^{-4}. The learning rate is divided by ten after 30 epochs. The training batch size is set to eight. During TTT stage, we set the learning rate to 1×10^{-4} and update model parameters for four gradient descent steps. All the input RSIs are resized to 256×256 before being fed into the model. Following common practices [7], [9], [21], our method is trained using augmented data, which include flipping and rotation of the original training data. This augmentation strategy generates seven additional variants of the original training data.

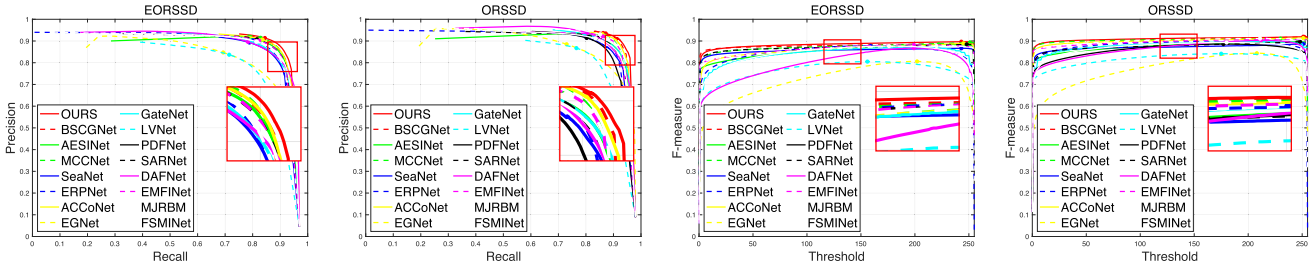
4) Comparison Methods: We conduct a comprehensive comparison of our method with 21 competitive methods, including five CNN-based NSI-SOD methods (i.e., DSS [30], RADF [64], EGNet [34], GateNet [31], PoolNet [36]), three traditional RSI-SOD methods (i.e., SMFF [65], VOS [66], CMC [67]), and 13 CNN-based RSI-SOD methods (i.e., LVNet [5], PDFNet [6], SARNet [8], DAFNet [7], EMFINet [11], MJRBM [45], FSMINet [68], MCCNet [10], SeaNet [69], ERPNet [12], ACCoNet [21], BSCGNet [70], and AESINet [71]). The detailed experimental comparison of these methods is presented in the following.

B. Experimental Comparison on Clean RSI Data

1) Quantitative Comparison: The quantitative comparison of our proposed method against other methods on the EORSSD and ORSSD datasets is presented in Table I. Our method exhibits superior performance, leading in five metrics on EORSSD and eight metrics on ORSSD. On ORSSD, our method improves the second-best method by 0.64%, 0.76%, 0.56%, and 15.05% in terms of F_β^{mean}, F_β^{ada}, F_β^{ada}, and MAE, respectively. On EORSSD, our method achieves consistent better performance than the second-best method in terms of S_m (0.9385 versus 0.9364), $F_\beta^{\max}$ (0.8900 versus 0.8861), F_β^{mean} (0.8745 versus 0.8664), and $E_\varepsilon^{\text{mean}}$ (0.9722 versus 0.9679). Moreover, we extend our comparison to the larger ORSI-4199 dataset, as shown in Table II. This dataset's extensive size offers a comprehensive platform to evaluate scalability, and the results affirm the effectiveness of our method on datasets of varying magnitudes. The PR and F-measure curves are presented in Fig. 6. Our method's curves are notably positioned above those of other methods, highlighting its exceptional capability in SOD across various thresholds. Despite our architecture's multibranch, multitask design, which results in a sizable parameter count of 81.26 **M**, it maintains a competitive balance in terms of FLOPs (226.0 **G**) and FPS (33.1). This places it in the middle range among deep-learning-based methods in terms of computational efficiency.

2) Qualitative Comparison: We show the visual comparison of our method and 11 representative state-of-the-art methods across four challenging scenarios in Fig. 7. 1) *Multiple Small Objects (First-Second Rows):* While most existing methods tend to overlook certain small objects or incur background clutter, our method consistently and clearly delineates all small objects. 2) *Intricate Object Topology (Third and Fourth Rows):* Our method excels in accentuating salient objects, maintaining their structural integrity. In contrast, other representative methods, such as AESINet and ACCoNet, struggle to capture the object structures. 3) *Strong Background Interference (Fifth and Sixth Rows):* The compared methods are adversely influenced by background interference, for example, object shadows in the fifth row and rural building in the sixth row. Our method effectively

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(a) (b) (c) (d)

Fig. 6. Visualization of PR and F-measure curves. (a) PR curve on EORSSD. (b) PR curve on ORSSD. (c) F-measure curve on EORSSD. (d) F-measure curve on ORSSD.

TABLE I

QUANTITATIVE EVALUATION ON THE EORSSD AND ORSSD DATASETS IN TERMS OF MAE (M), STRUCTURE MEASURE (S_m), MAXIMUM, MEAN, AND ADAPTIVE F-MEASURE ($F_\beta^{\max}$, F_β^{mean}, F_β^{ada}), MAXIMUM, MEAN, AND ADAPTIVE E-MEASURE ($E_\varepsilon^{\max}$, $E_\varepsilon^{\text{mean}}$, $E_\varepsilon^{\text{ada}}$). THE SYMBOLS ↑ / ↓ INDICATE A LARGER/SMALL VALUE IS BETTER. THE TOP THREE RESULTS ARE MARKED IN RED, GREEN, AND BLUE, RESPECTIVELY

Method	Param (M)↓	FLOPs (G)↓	Speed (FPS)↑	EORSSD								ORSSD							
				S_m↑	M↓	F_β^{max}↑	F_β^{mean}↑	F_β^{ada}↑	E_ε^{max}↑	E_ε^{mean}↑	E_ε^{ada}↑	S_m↑	M↓	F_β^{max}↑	F_β^{mean}↑	F_β^{ada}↑	E_ε^{max}↑	E_ε^{mean}↑	E_ε^{ada}↑
DSS	-	-	-	.7868	.0186	.6849	.5801	.4597	.9186	.7631	.6933	.8262	.0363	.7467	.6962	.6202	.8860	.8362	.8085
RADF	-	-	-	.8179	.0168	.7446	.6582	.4933	.9140	.8567	.7261	.8259	.0382	.7619	.6856	.5730	.9130	.8298	.7678
EGNet	108.1	313.9	43.5	.8601	.0110	.7880	.6967	.5379	.9570	.8775	.7566	.8721	.0216	.8332	.7500	.6452	.9731	.9013	.8226
GateNet	100.0	96.3	51.5	.9114	.0095	.8566	.8228	.7109	.9610	.9385	.8909	.9186	.0137	.8871	.8679	.8229	.9664	.9538	.9428
PoolNet	52.5	97.6	80.9	.8207	.0210	.7545	.6406	.4611	.9292	.8193	.6836	.8403	.0358	.7706	.6999	.6166	.9343	.8650	.8124
CMC	-	-	-	.5798	.1057	.3268	.2692	.2007	.6803	.5894	.4890	.6033	.1267	.3913	.3454	.3108	.7064	.6417	.5996
SMFF	-	-	-	.5401	.1434	.5176	.2992	.2083	.7744	.5197	.5014	.5312	.1854	.4417	.2684	.2496	.7402	.4920	.5676
VOS	-	-	-	.5082	.2096	.2765	.2107	.1836	.5982	.4886	.4767	.5366	.2151	.3471	.2717	.2633	.6514	.5352	.5826
LVNet	207.0	-	1.5	.8630	.0146	.7794	.7328	.6284	.9254	.8801	.8445	.8815	.0207	.8263	.7995	.7506	.9456	.9259	.9195
PDFNet	228.0	-	-	-	-	-	-	-	-	-	-	.9112	.0149	.8726	.8435	.7924	.9608	.9403	.9341
SARNet	25.9	137.3	43.4	.9182	.0115	.8695	.8373	.8030	.9597	.9514	.9431	.9048	.0206	.8733	.8489	.8346	.9511	.9436	.9405
DAFNet	29.4	548.1	42.5	.9166	.0060	.8612	.7842	.6423	.9859	.9290	.8443	.9191	.0113	.8928	.8511	.7876	.9771	.9539	.9360
EMFINet	95.1	353.1	18.6	.9290	.0084	.8720	.8486	.7984	.9711	.9604	.9501	.9366	.0109	.9002	.8856	.8617	.9737	.9671	.9663
MJRBM	43.8	101.3	15.5	.9197	.0099	.8656	.8239	.7066	.9646	.9350	.8897	.9204	.0163	.8842	.8566	.8022	.9623	.9415	.9328
FSMINet	3.6	9.7	17.7	.9097	.0120	.8481	.8293	.7963	.9524	.9408	.9426	.9087	.0161	.8656	.8525	.8413	.9564	.9458	.9501
MCCNet	67.7	234.2	39.5	.9317	.0074	.8842	.8618	.8238	.9733	.9679	.9557	.9376	.0109	.9053	.8933	.8795	.9723	.9682	.9660
SeaNet	2.7	2.6	107.6	.9150	.0083	.8560	.8435	.8184	.9666	.9602	.9559	.9197	.0124	.8786	.8599	.8483	.9643	.9595	.9587
ERPNet	56.5	227.4	73.7	.9172	.0098	.8629	.8275	.7349	.9615	.9310	.9125	.9182	.0160	.8832	.8624	.8278	.9601	.9406	.9465
ACCoNet	102.6	368.5	59.9	.9364	.0070	.8861	.8591	.8103	.9732	.9642	.9501	.9392	.0119	.9040	.8824	.8639	.9722	.9646	.9646
BSCGNet	27.0	172.8	-	.9294	.0079	.8807	.8664	.8504	.9692	.9649	.9631	.9411	.0093	.9101	.8978	.8897	.9784	.9745	.9737
AESINet	41.45	110.6	35.8	.9314	.0076	.8749	.8420	.7811	.9706	.9588	.9410	.9222	.0135	.8857	.8613	.8254	.9647	.9550	.9452
OURS	81.26	226.0	33.1	.9385	.0064	.8900	.8745	.8440	.9768	.9722	.9642	.9465	.0079	.9171	.9036	.8947	.9824	.9784	.9811

TABLE II

QUANTITATIVE EVALUATION ON THE ORSI-4199 DATASET. THE BEST RESULTS ARE MARKED IN BOLD

Method	S_m↑	M↓	$E_\varepsilon^{\max}$↑	$F_\beta^{\max}$↑
ERPNet	.7867	.0663	.8683	.7594
DAFNet	.7464	.0810	.8314	.7019
EMFINet	.8688	.0334	.9339	.8588
MJRBM	.8593	.0374	.9311	.8493
FSMINet	.8614	.0362	.9282	.8468
MCCNet	.8686	.0338	.9346	.8592
SeaNet	.8634	.0345	.9357	.8562
ACCoNet	.8727	.0314	.9370	.8631
AESINet	.8656	.0343	.9379	.8629
Ours	**.8756**	**.0308**	**.9409**	**.8689**

mitigates such background noise, yielding clean saliency maps. 4) *Large Object Regions (Seventh and Eighth Rows):* Most existing methods fail to uniformly highlight the entire salient regions. In contrast, our method not only captures the entire salient regions but also preserves their boundary structures. The remarkable capability in preserving object structures and details can be attributed to synergistic integration of multitask learning and test-time adaptation in our proposed method.

C. Experimental Comparison on Corrupted RSI Data

1) Quantitative Comparison: We present quantitative comparison of our proposed method against other deep-learning-based RSI-SOD methods on the corrupted versions of the EORSSD and ORSSD datasets in Table III. These corrupted data are generated using three typical corruptions, i.e., GB, GN, and CR, to simulate realistic, challenging remote sensing imaging conditions. We present two variants of our method in the table. The first is the entire architecture with test-time adaptation (denoted as OURS-TT). The second is the entire architecture without test-time adaptation, i.e., with only multitask joint training (denoted as OURS-JT). It can

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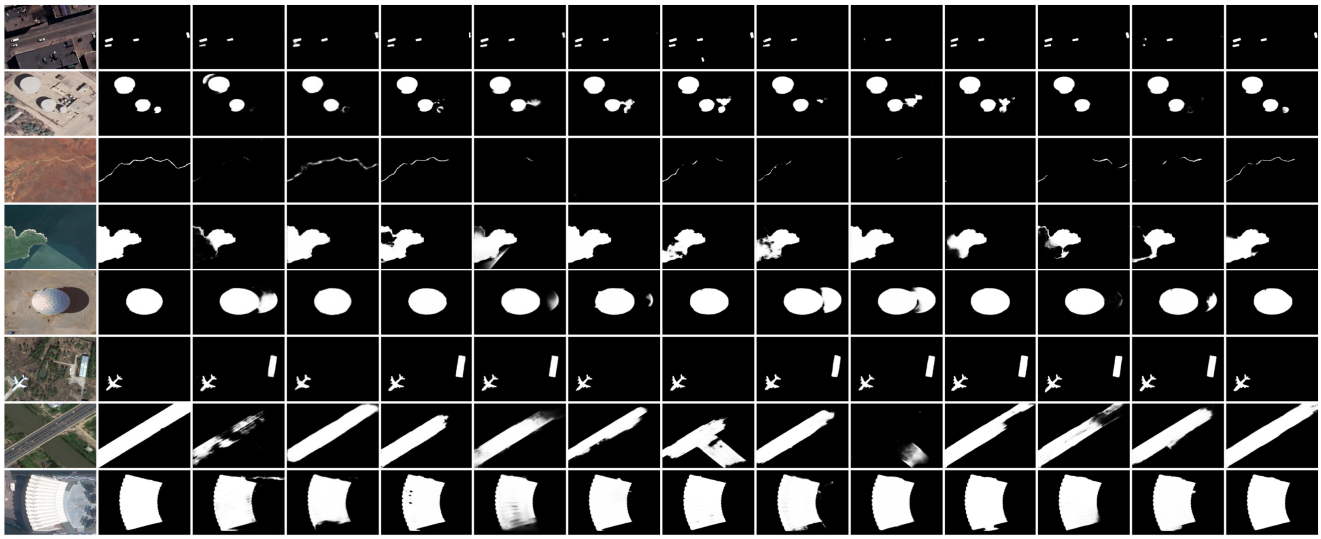


Fig. 7. Visual comparison of our method against other methods on normal RSI data.

TABLE III

QUANTITATIVE EVALUATION ON THE CORRUPTED EORSSD AND ORSSD DATASETS IN TERMS OF MAE (M), STRUCTURE MEASURE (S_m), MAXIMUM, MEAN, AND ADAPTIVE F-MEASURE ($F_\beta^{\max}$, F_β^{mean}, F_β^{ada}), MAXIMUM, MEAN, AND ADAPTIVE E-MEASURE ($E_\varepsilon^{\max}$, $E_\varepsilon^{\text{mean}}$, AND $E_\varepsilon^{\text{ada}}$). THE SYMBOLS ↑/↓ INDICATE A LARGER/SMALL VALUE IS BETTER. THE BEST RESULTS ARE MARKED IN BOLD

Method	C.T.	EORSSD								ORSSD							
		$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{ada}\uparrow$	$E_\varepsilon^{max}\uparrow$	$E_\varepsilon^{mean}\uparrow$	$E_\varepsilon^{ada}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$F_\beta^{mean}\uparrow$	$F_\beta^{ada}\uparrow$	$E_\varepsilon^{max}\uparrow$	$E_\varepsilon^{mean}\uparrow$	$E_\varepsilon^{ada}\uparrow$
EMFINet	GN	.8896	.0104	.8174	.8008	.7813	.9303	.9081	.9297	.8885	.0216	.8380	.8267	.8225	.9325	.9148	.9356
DAFNet	GN	.7823	.0239	.6517	.6131	.5473	.8347	.7874	.7868	.7446	.0467	.6053	.5808	.5940	.7764	.7321	.7911
FSMINet	GN	.8730	.0138	.8050	.7908	.7758	.9313	.9049	.9303	.8618	.0268	.7978	.7869	.7801	.9163	.8960	.9183
MCCNet	GN	.8735	.0166	.7973	.7864	.7528	.9100	.9001	.9110	.8595	.0311	.8013	.7957	.7943	.9033	.8816	.9025
SeaNet	GN	.8812	.0146	.8118	.8017	.7957	.9364	.9220	.9357	.8734	.0270	.8195	.8114	.8167	**.9309**	.9072	**.9332**
ERPNet	GN	.8809	.0156	.8005	.7704	.7064	.9201	.8926	.9026	.8199	.0384	.7403	.7159	.7099	.8715	.8185	.8735
ACCoNet	GN	.8941	.0130	.8268	.8125	.7872	.9394	.9213	.9373	.8826	.0253	.8238	.8151	.8149	.9289	.9001	.9291
AESINet	GN	.8871	.0132	.8145	.7915	.7489	.9254	.9134	.9155	.8657	.0257	.7915	.7773	.7662	.8939	.8818	.8937
OUES-JT	GN	.8934	.0142	.8241	.8152	.8007	.9360	.9260	.9383	.8701	.0273	.8083	.8028	.8012	.9007	.8877	.9034
OURS-TT	GN	**.9111**	**.0124**	**.8519**	**.8382**	**.8159**	**.9521**	**.9474**	**.9462**	**.8934**	**.0246**	**.8446**	**.8364**	**.8345**	.9299	**.9257**	.9323
EMFINet	GB	.8682	.0149	.7894	.7586	.7233	.9189	.9095	.9045	.8637	.0281	.8055	.7839	.7749	.9100	.9059	.9223
DAFNet	GB	.8423	.0245	.7750	.7464	.7070	.9064	.8752	.8780	.8302	.0144	.7385	.6954	.5994	.9113	.8670	.8217
FSMINet	GB	.8302	.0235	.7251	.7125	.6959	.8914	.8661	.8899	.8627	.0323	.7936	.7791	.7598	.9139	.9022	.9102
MCCNet	GB	.8418	.0184	.7466	.7208	.6897	.8908	.8862	.8799	.8303	.0369	.7431	.7309	.7322	.8815	.8645	.8905
SeaNet	GB	.8523	.0164	.7498	.7390	.7360	.9151	.8975	.9140	.8611	.0225	.7944	.7862	.7896	**.9289**	.8983	.9248
ERPNet	GB	.8439	.0200	.7391	.7064	.6454	.8778	.8651	.8608	.7725	.0600	.6655	.6461	.6202	.8267	.8053	.8504
ACCoNet	GB	.8651	.0158	.7854	.7470	.7039	.9170	.9069	.8953	.8601	.0270	.7906	.7709	.7603	.9104	.9006	.9182
AESINet	GB	**.9052**	**.0105**	**.8410**	**.8047**	**.7467**	**.9525**	**.9404**	**.9195**	**.8785**	.0236	.8145	.7943	.7756	.9128	.9046	.9038
OUES-JT	GB	.8572	.0171	.7694	.7453	.7328	.9088	.9038	.9028	.8430	.0291	.7585	.7486	.7605	.9022	.8705	.9051
OURS-TT	GB	.8664	.0150	.7888	.7593	.7336	.9197	.9151	.9098	.8723	**.0218**	**.8173**	**.7970**	**.7938**	.9237	**.9203**	**.9261**
EMFINet	CR	.8437	.0193	.7243	.7117	.6979	.8788	.8438	.8741	.8234	.0424	.7466	.7266	.7444	.8706	.8282	.8795
DAFNet	CR	.7727	.0213	.6429	.5939	.5599	.8367	.7576	.8003	.7974	.0389	.7067	.6754	.6802	.8655	.8098	.8604
FSMINet	CR	.6249	.0474	.3591	.3179	.3609	.6631	.5539	.6540	.7633	.0502	.6121	.5977	.6148	.8307	.7497	.8359
MCCNet	CR	.8575	.0183	.7645	.7583	.7434	.9042	.8794	.9029	.7568	.0511	.6698	.6288	.6857	.8096	.7291	.8189
SeaNet	CR	.8165	.0192	.7040	.6853	.7085	.8744	.8259	.8777	.8202	**.0325**	.7411	.7217	.7455	**.8999**	.8325	**.8911**
ERPNet	CR	.8268	.0270	.7079	.6776	.6401	.8662	.8079	.8481	.7192	.0960	.5685	.5395	.5326	.7452	.7029	.7792
ACCoNet	CR	.8732	.0157	.7806	.7677	.7360	.9190	.8865	.9096	.8410	.0386	.7569	.7450	.7619	.8825	.8449	.8977
AESINet	CR	.8613	.0162	.7716	.7563	.7400	.8974	.8719	.8974	.7554	.0475	.6037	.5957	.6165	.7491	.7258	.7563
OUES-JT	CR	.8820	.0164	.7984	.7924	.7848	.9188	.8998	.9247	.8237	.0341	.7281	.7177	.7544	.8702	.8164	.8732
OURS-TT	CR	**.8940**	**.0151**	**.8232**	**.8139**	**.7993**	**.9308**	**.9235**	**.9346**	**.8502**	.0355	**.7763**	**.7705**	**.7701**	.8803	**.8651**	.8766

* C.T.: RSI Corruption Type, GB: Gaussian Blur, GN: Gausisan Noise, CR: Contrast Reduction.

be observed that OURS-TT demonstrates consistent superior performance to other compared methods. On the EORSSD corrupted with GN, OURS-TT achieves the best results in all the eight metrics, including an impressive S_m of 0.9111 and $F_\beta^{\max}$ of 0.8519. When applied to the ORSSD under GN, OURS-TT maintains its superior performance by ranking first in six metrics. Our method's robustness is further exemplified in scenarios with GB and CR. On the EORSSD corrupted with GB, AESINet surpasses our method and other compared methods, which may be attributed to its capability of sensing subtle difference when detecting object edges and details. The performance of OURS-TT improves steadily based on

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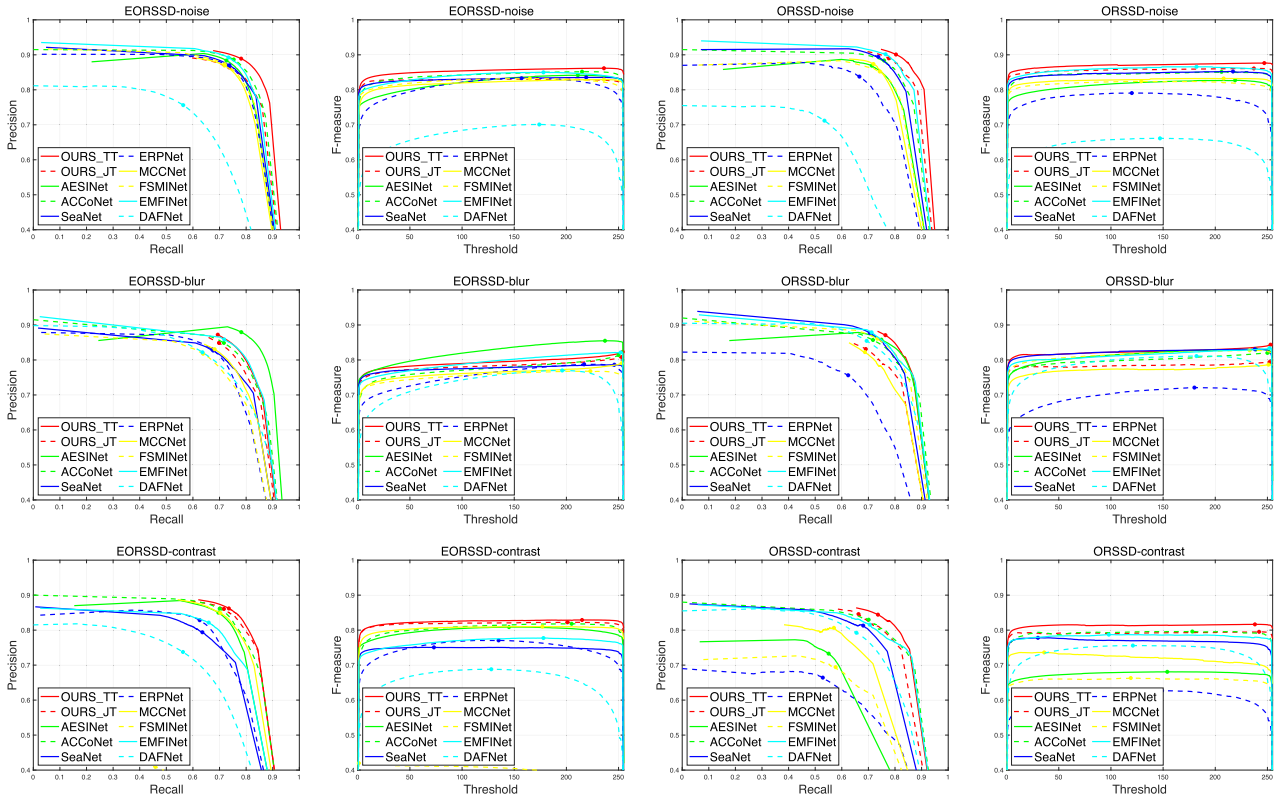


Fig. 8. Visualization of PR and F-measure curves on corrupted RSI data. The first rows show the curves on data corrupted with GN. The second rows show the curves on data corrupted with GB. The third rows show the curves on data corrupted with CR.

OURS-JT, demonstrating the effectiveness and necessity of the test-time adaptation. Corrupted RSIs introduce a distribution shift from the clean RSI data used for training models. Since most existing methods rely on a supervised training paradigm without strategies to address this distribution shift, they struggle to perform effectively on corrupted RSIs. The PR and F-measure curves on corrupted RSI data are presented in Fig. 8. It is observed that our method outperforms other methods in overall, demonstrating that our method is robust in detecting salient objects in varying thresholds.

2) Qualitative Comparison: We present the visual comparison of our method against other deep-learning-based RSI-SOD methods on corrupted RSI datasets in Fig. 9. 1) *GB Corruption (First–Third Rows):* Most methods tend to discard certain object parts or small objects due to visual blurriness. In comparison, our method can highlight salient regions uniformly when detecting objects of varying scales. 2) *GN Corruption (Fourth–Sixth Rows):* While other outstanding methods, such as ACCoNet and AESINet, are negatively influenced by the noise, our method can still pop out the entire salient regions clearly. 3) *CR (Seventh–Ninth Rows):* It is challenging and difficult for the existing methods to highlight constituent salient regions due to the lack of contrast. Nevertheless, our method shows remarkable performance in tackling this challenge in various situations, including intricate topology and multiple tiny objects. In addition, OURS-TT improves OURS-JT substantially in visual quality, demonstrating the effectiveness of the test-time adaptation strategy adopted in our method.

D. Ablation Study

1) Individual Contribution of the Key Components in Our Approach: We conduct ablation experiments to validate the effectiveness of each key component (or step) in our method on both clean and corrupted ORSSD datasets. The ablation results are presented in Table IV. "BL" denotes the U-shaped baseline with only the shared encoder and primary-task decoder. "AS" denotes ASPP module that is integrated into the decoding stage of the primary-task decoder. "AD" denotes the auxiliary-task decoder for RSI reconstruction. "CM" denotes the CMM. "TT" denotes test-time training step that is performed before the final inference. It is observed that the baseline achieves only a moderate performance. With the ASPP, the performance is improved due to the thorough multiscale context modeling capability. As depicted in the third row, multitask learning by incorporating the auxiliary-task decoder contributes to performance improvement. Notably, as shown in the fourth row, cross-task modulation can reach a remarkable performance gain on both the clean and corrupted data, demonstrating the necessity of collaboration between the dual-stream decoders. The fifth and sixth rows exhibit the results of applying TTT before the final inference. It can be observed that TTT has minor influence on the clean RSI data, while has a significant impact on the corrupted data. It justifies the importance of TTT in dealing with severe image condition degradation in remote sensing scenarios. Our final architecture with TTT achieves an excellent performance in various situations, showing high robustness and generalization capability.

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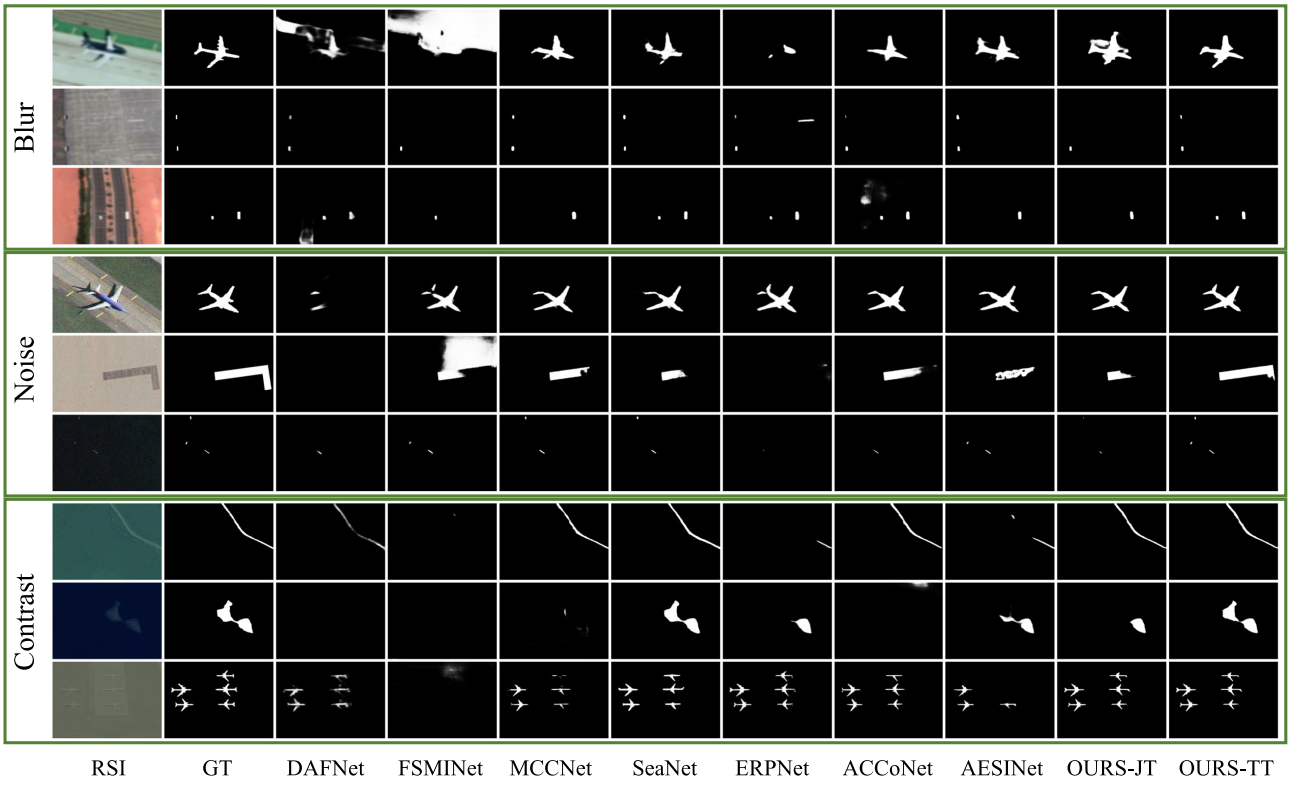


Fig. 9. Visual comparison of our method against other methods on the corrupted RSI data.

TABLE IV
ABLATION STUDY ON THE KEY COMPONENTS OF OUR PROPOSED APPROACH

BL	AS	AD	CM	TT	Clean			Blur			Noise			Contrast		
					$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$
✓					.9296	.0131	.8917	.7694	.0387	.6323	.7836	.0421	.6987	.7151	.0503	.5748
✓	✓				.9358	.0125	.9042	.8023	.0445	.6945	.8480	.0331	.7962	.7775	.0484	.6793
✓	✓	✓			.9396	.0087	.9098	.7953	.0359	.6951	.8231	.0357	.7494	.7839	.0393	.6923
✓	✓	✓	✓		**.9467**	**.0077**	**.9176**	.8430	.0291	.7585	.8701	.0273	.8083	.8237	**.0341**	.7281
✓	✓	✓		✓	.9375	.0092	.9110	.8348	.0288	.7381	.8778	.0292	.8169	.8111	.0343	.7248
✓	✓	✓	✓	✓	.9465	.0079	.9171	**.8723**	**.0218**	**.8173**	**.8934**	**.0246**	**.8446**	**.8502**	.0355	**.7763**

2) Effectiveness of Each Component in CMM: We present ablation experiments to examine the effectiveness of each component within CMM and report them in Table V. The experiments are carried out on clean datasets. We investigate four variants of CMM, including sequential space and channel modulation ("sequential mod."), parallel space and channel modulation ("parallel mod."), space-only modulation ("only space mod."), and channel-only modulation ("only channel mod."). Our final architecture adopts the sequential modulation, which performs slightly better than the parallel modulation and other variants. As shown in the third and fourth rows, using only one type of modulation decreases detection accuracy, demonstrating that both space and channel modulation are necessary. The last row indicates replacing the CMM with a simple addition operation for cross-task feature modulation ("w/o CMM"). Notably, the performance degrades severely without the CMM module, justifying its deployment in SOD.

TABLE V
ABLATION STUDY ON THE CROSS-TASK MODULATION MODULE (CMM)

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
Sequential Mod.	**.9465**	**.0079**	**.9171**	**.9385**	.0064	**.8900**
Parallel Mod.	.9435	.0091	.9148	.9360	.0067	.8883
Only Channel Mod.	.9443	.0093	.9166	.9371	**.0062**	.8900
Only Space Mod.	.9453	.0084	.9154	.9349	.0069	.8871
w/o CMM	.9385	.0101	.9080	.9322	.0070	.8850

3) Comparison to Other TTT Approaches: We compare our method with other prevalent TTT approaches, including TTT-Rot [46], TENT [47], and UDA [72]. The quantitative evaluation is performed on corrupted EORSSD dataset, and the results are reported in Table VI. We have tailored these TTT methods to fit our architecture for fair comparison. For evaluating TTT-Rot, we set the auxiliary-task decoder to predict

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TABLE VI
COMPARISON OF OUR METHOD WITH SEVERAL OTHER TTT APPROACHES. THE PERFORMANCE IS EVALUATED ON CORRUPTED EORSSD

Method	Blur				Noise				Contrast			
	$S_m\uparrow$	$M\downarrow$	$E_\varepsilon^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$E_\varepsilon^{\max}\uparrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$E_\varepsilon^{\max}\uparrow$	$F_\beta^{\max}\uparrow$
TTT-Rot	.8504	.0165	.8985	.7577	.8731	.0159	.9088	.7877	.8711	**.0142**	.9061	.7823
TENT	.7705	.0155	.7105	.5459	.7986	.0139	.7385	.5858	.7845	.0156	.7157	.5550
UDA	.8405	.0167	.8836	.7376	.9018	.0130	.9490	.8426	.8757	.0144	.9108	.7887
OURS	**.8664**	**.0150**	**.9197**	**.7888**	**.9111**	**.0124**	**.9521**	**.8519**	**.8940**	.0151	**.9308**	**.8232**

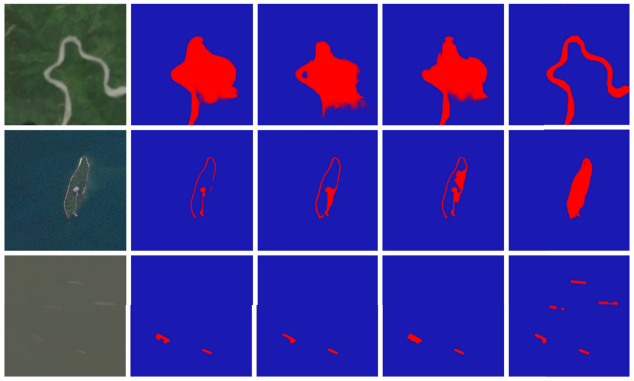


Fig. 10. Qualitative comparison of different test-time adaptation methods. (a) Optical RSI, first row: GB, second row: GN, and third row: CR. (b) UDA [72]. (c) TTT-Rot [46]. (d) TENT [47]. (e) Our method.

image rotations and remove the CMM. The architecture is trained using SOD and rotation prediction simultaneously and adjusted by the rotation prediction task in test time. For evaluating TENT and UDA, we keep the primary encoder–encoder architecture and remove the auxiliary-task decoder. TENT adjusts the batch normalization layers by minimizing the BCE of the model predictions. UDA adapts the running mean and variance of the batch normalization by the statistics of the incoming test sample. Table VI shows that our proposed TTT method achieves superior performance to the compared TTT methods. It may be attributed to that our TTT method is designed specifically for pixel-level representation learning and can benefit more on SOD task. Fig. 10 displays heatmaps of different test-time adaptation methods on corrupted RSIs. It is evident that other methods have imperfections and struggle to adapt to degraded remote sensing scenes. As illustrated in the first row, image blur causes confusion between object regions and background areas. As illustrated in the third row, extremely low contrast in remote sensing scenes poses significant difficulties and challenges for object discrimination. In contrast, our TTT method demonstrates superior adaptability in these challenging conditions, consistently highlighting objects clearly and uniformly.

VI. CONCLUSION

In this article, we proposed a novel multitask collaboration approach to improve model's test-time adaptability for RSI-SOD using only test data without labels. Our approach integrates a self-supervised auxiliary task, i.e., image reconstruction, with the supervised primary task, i.e., saliency prediction, to achieve collaboration learning, by an architecture comprising a shared feature encoder and two separate task-specific decoders. In addition, we designed a CMM to exploit intertask correlation for enriching saliency representations. During training, the task-specific decoders are optimized simultaneously to extract universal and comprehensive feature representations. At inference, the self-supervised auxiliary task decoder is further optimized unlabeled test data, allowing the model to adapt to test distribution. Extensive experimental evaluations demonstrate the superiority of our method to other state-of-the-art methods and the robustness of the proposed test-time adaptation strategy across various testing scenarios. Future research may focus on developing more sophisticated models to enhance test-time adaptability and extending them to multimodal images.

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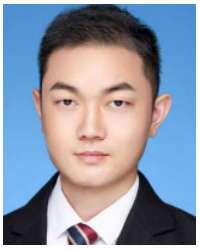
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